

Q1. Motivation and Contract Terms

The client locks in today the price at which it will purchase the 10-year US Treasury one year from now, hedging future price/rate risk (or expressing a view that rates will fall and the bond price will rise).

Key term-sheet parameters:

- Notional: \$1,000,000
- Underlying bond (coupon, maturity T_m)
- Settlement date T_s
- Forward price / strike K (set at inception so the contract has zero value)
- Physical vs. cash settlement
- Treatment of accrued interest/coupons
- Collateral / CSA terms

At T_s , under physical settlement, Bank A delivers the bond to the client and the client pays K ; the client's payoff is $V(T_s) = B(T_s; T_s, T_m) - K$.

Q2. Hedging the Spot Position

The desk buys the bond spot at $t = 0$ so it is immediately flat to the underlying bond price, any subsequent move in $B(t; t, T_m)$ is offset because the desk already owns the bond it must deliver at T_s . Without this, the desk would be short the forward and exposed to market risk during $[0, T_s]$.

To fund the purchase, the desk does an internal repo trade: it posts the bond as collateral to the repo desk and borrows cash at the term repo rate R ; the treasury desk facilitates/prices this internal funding. Economically this replicates a cash and carry: long bond, financed by repo.

Q3. Computing $B(0; T_s, T_m)$

Borrowing \$1 from the repo desk for the term requires repaying $1 + RT_s$ at T_s (simple interest, term T_s in years).

Buying the bond costs $B(0; 0, T_m)$, so the desk owes $B(0; 0, T_m)(1 + RT_s)$ at T_s . Coupons c_i received at times $t_i \in (0, T_s]$ (while the bond is posted as collateral) are used to pay down the repo debt, and each is future-valued to T_s at the repo rate:

$$B(0; T_s, T_m) = B(0; 0, T_m) (1 + RT_s) - \sum_{i: t_i \leq T_s} c_i (1 + R(T_s - t_i)).$$

This is the standard cost-of-carry forward price: spot grown at the financing rate, less the future value of coupons earned during the carry period.

Q4. Marking to Market at $t \in (0, T_s]$

Recompute the forward price using the then-current spot bond price and the remaining repo carry to T_s :

$$B(t; T_s, T_m) = B(t; t, T_m) (1 + R'(T_s - t)) - \sum_{i: t < t_i \leq T_s} c_i (1 + R'(T_s - t_i)),$$

where R' is the applicable repo rate for the remaining term. The MTM is then

$$V(t) = df^{csa}(t, T_s) (B(t; T_s, T_m) - K).$$

Q5. Risk and Revenue

Because the desk is long the bond and short the forward, it is hedged against outright bond price risk. It is not, however, hedged against repo risk: the repo rate actually realized (or the "specialness" of this bond in the repo market, since it may repo below general collateral rates) can differ from what was assumed when quoting K . If repo is done on a rolled overnight basis rather than a locked term rate, there is also rollover risk.

So the desk isn't risk-free on Day 1 economics, and to earn revenue it builds a spread into K : charging the client a slightly worse repo-implied forward price than the desk's own funding rate to compensate for repo/ specialness risk, funding costs, capital usage, and operational/credit risk.

Q5*. Connection to Risk-Neutral Pricing

The cash-and-carry replication (long bond, financed at the repo rate) is economically the same as pricing under the risk-neutral measure with the repo-rate money-market account as numeraire: the repo financing leg plays the role of the risk-free discounting/growth factor, and the forward price $B(0; T_s, T_m)$ is exactly the time- T_s expected value of the bond under that measure, net of the (deterministic) coupon carry. Under the assumption that the repo rate is deterministic (or at least a valid financing/discount rate), this replicating-portfolio argument coincides with the change-of-numeraire/martingale pricing result.